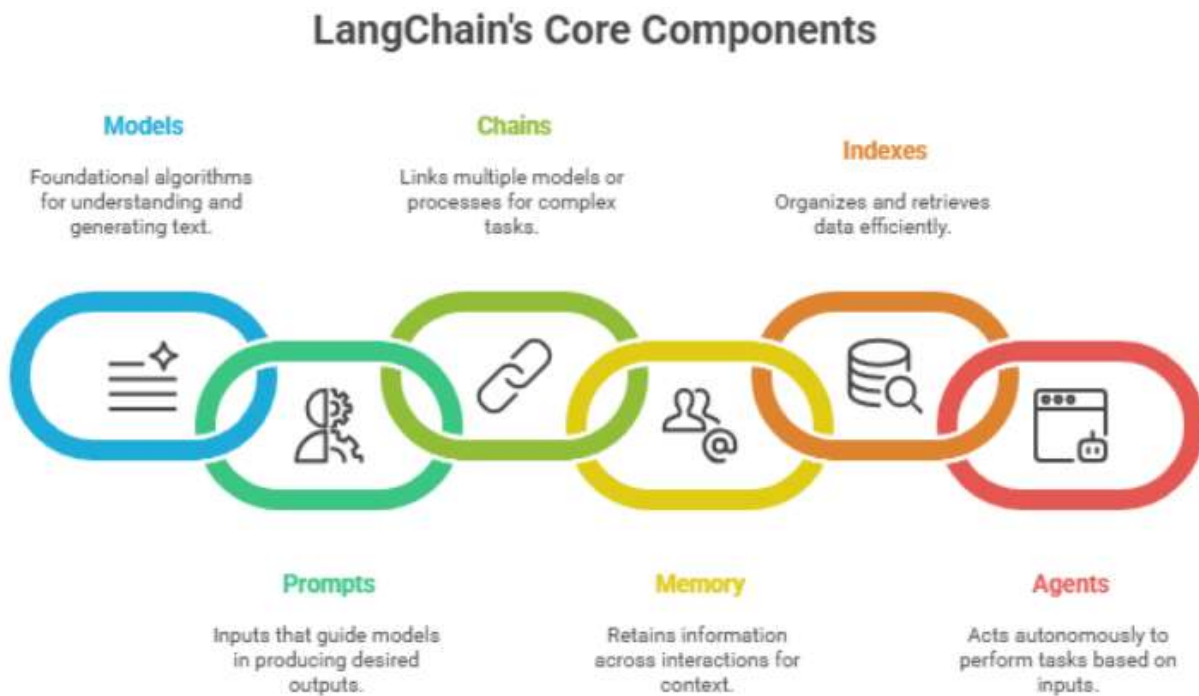


Langchain

LangChain is an open-source framework used to build agents and applications powered by Large Language Models (LLMs).

Components of Langchain:



1. Models

Models are the core intelligence of a **LangChain** application.

Types:

- **LLMs (Large Language Models)** → generate text from prompts
- **Chat Models** → conversation-based models
- **Embedding Models** → convert text into vectors

2. Prompts (Prompt Templates)

Prompts define how instructions are given to the model.

Instead of writing hardcoded text repeatedly, **LangChain** uses templates.

3. Chains

Chains connect multiple components into a sequence.

Flow:

Input → Prompt → LLM → Output

4. Memory

Memory stores previous interactions.

Without memory:

```
User: My name is Ali
User: What is my name?
Bot: I don't know
```

With memory:

```
Bot: Your name is Ali
```

Purpose:

- Maintain conversation context
- Personalize chatbots

5. Indexes

a. Document Loaders

Load data from different sources.

Examples:

- PDF
- Website
- Text files
- CSV
- Word files

b. Text Splitters

Large documents must be split into smaller chunks because LLMs have token limits.

c. Vector Stores

Store embedding vectors and allow similarity search.

Popular vector databases:

- ChromaDB
- FAISS
- Pinecone
- Weaviate

d. Retrievers

Retrievers search the vector database and return relevant documents.

Purpose:

- Fetch useful context
- Used heavily in RAG systems

6. Agents

Agents allow the model to decide which tools to use.

- Instead of a fixed flow:
- Input → Output
- **Agent:**
- Input
 - ↓
 - Reason
 - ↓
 - Choose Tool
 - ↓
 - Execute
 - ↓
 - Answer